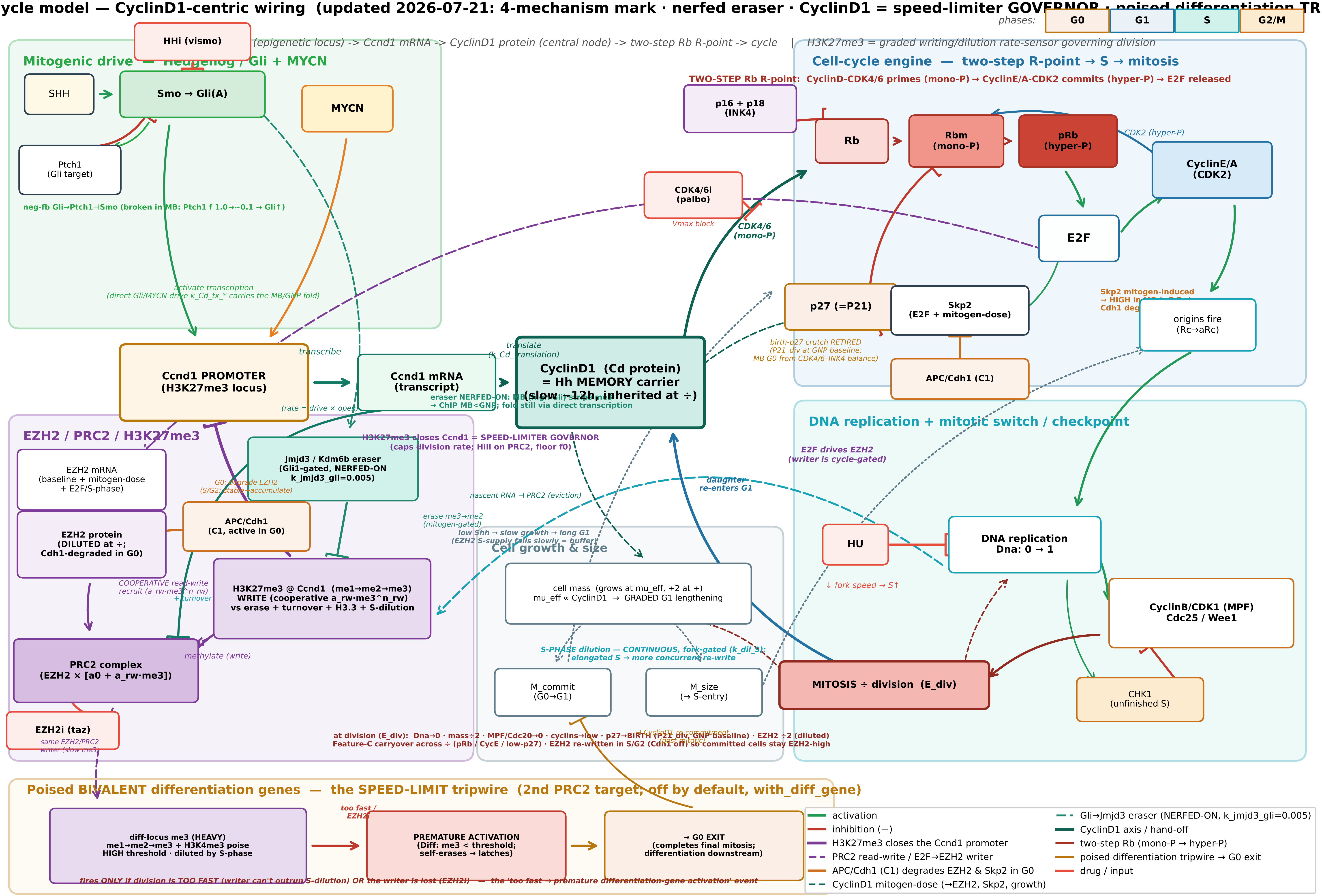
v44 cell-cycle model — CyclinD1-centric wiring (updated 2026-07-21: 4-mechanism mark · nerfed eraser · CyclinD1 = speed-limiter GOVERNOR · poised differentiation TRIPWIRE)



(epigenetic locus) -> Ccnd1 mRNA -> CyclinD1 protein (central node) -> two-step Rb R-point -> cycle | H3K27me3 = graded writing/dilution rate-sensor governing division

phases: G0 | G1 | S | G2/M

Mitogenic drive — Hedgehog / Gli + MYCN

HHi (vismo)

SHH

Smo → Gli(A)

MYCN

Ptch1 (Gli target)

neg-fb Gli→Ptch1⊣Smo (broken in MB: Ptch1 f 1.0→~0.1 → Gli↑)

activate transcription (direct Gli/MYCN drive k_Cd_tx_* carries the MB/GNP fold)

Ccnd1 PROMOTER (H3K27me3 locus)

(rate = drive × open)

transcribe

Ccnd1 mRNA (transcript)

translate (k_Cd_translation)

CyclinD1 (Cd protein) = Hh MEMORY carrier (slow ~12h, inherited at ÷)

eraser NERFED-ON: MB (high Gli) strips me3 → ChIP MB<GNP; fold still via direct transcription

H3K27me3 closes Ccnd1 = SPEED-LIMITER GOVERNOR (caps division rate; Hill on PRC2, floor f0)

CDK4/6i (palbo)

Vmax block

CDK4/6 (mono-P)

Cell-cycle engine — two-step R-point → S → mitosis

TWO-STEP Rb R-point: CyclinD-CDK4/6 primes (mono-P) → CyclinE/A-CDK2 commits (hyper-P) → E2F released

p16 + p18 (INK4)

Rb

Rbm (mono-P)

pRb (hyper-P)

CDK2 (hyper-P)

CyclinE/A (CDK2)

E2F

p27 (=P21)

Skp2 (E2F + mitogen-dose)

APC/Cdh1 (C1)

birth-p27 crutch RETIRED (P21_div at GNP baseline; MB G0 from CDK4/6–INK4 balance)

Skp2 mitogen-induced
→ HIGH in
Cdh1 deg

origins fire (Rc→aRc)

EZH2 / PRC2 / H3K27me3

EZH2 mRNA (baseline + mitogen-dose + E2F/S-phase)

EZH2 protein (DILUTED at ÷; Cdh1-degraded in G0)

APC/Cdh1 (C1, active in G0)

G0: degrade EZH2 (S/G2: stable→accumulate)

Jmjd3 / Kdm6b eraser (Gli1-gated, NERFED-ON k_jmjd3_gli=0.005)

erase me3→me2 (mitogen-gated)

COOPERATIVE read-write recruit (a_rw·me3^n_rw) + turnover

H3K27me3 @ Ccnd1 (me1→me2→me3) WRITE (cooperative a_rw·me3^n_rw) vs erase + turnover + H3.3 + S-dilution

methylate (write)

PRC2 complex (EZH2 × [a0 + a_rw·me3])

EZH2i (taz)

same EZH2/PRC2 writer (slow me3)

nascent RNA ⊣ PRC2 (eviction)

low Shh → slow growth → long G1 (EZH2 S-supply falls slowly = buffer)

daughter re-enters G1

E2F drives EZH2 (writer is cycle-gated)

Cell growth & size

cell mass (grows at mu_eff, ÷2 at ÷)
mu_eff ∝ CyclinD1 → GRADED G1 lengthening

S-PHASE dilution — CONTINUOUS, fork-gated (k_dil_S); elongated S → more concurrent re-write

M_commit (G0→G1)

M_size (→ S-entry)

⊣ CyclinD1 re-commitment (post-mitotic)

at division (E_div): Dna→0 · mass÷2 · MPF/Cdc20→0 · cyclins→low · p27→BIRTH (P21_div, GNP baseline) · EZH2 ÷2 (diluted)
Feature-C carryover across ÷ (pRb / CycE / low-p27) · EZH2 re-written in S/G2 (Cdh1 off) so committed cells stay EZH2-high

DNA replication + mitotic switch / checkpoint

HU

↓ fork speed → S↑

DNA replication Dna: 0 → 1

CyclinB/CDK1 (MPF) Cdc25 / Wee1

MITOSIS ÷ division (E_div)

CHK1 (unfinished S)

Poised BIVALENT differentiation genes — the SPEED-LIMIT tripwire (2nd PRC2 target; off by default, with_diff_gene)

diff-locus me3 (HEAVY) me1→me2→me3 + H3K4me3 poise HIGH threshold · diluted by S-phase

too fast / EZH2i

PREMATURE ACTIVATION (Diff: me3 < threshold; self-erases → latches)

→ G0 EXIT (completes final mitosis; differentiation downstream)

fires ONLY if division is TOO FAST (writer can't outrun S-dilution) OR the writer is lost (EZH2i) — the 'too fast → premature differentiation-gene activation' event

activation	Gli→Jmjd3 eraser (NERFED-ON, k_jmjd3_gli=0.005)
inhibition (⊣)	CyclinD1 axis / hand-off
H3K27me3 closes the Ccnd1 promoter	two-step Rb (mono-P → hyper-P)
PRC2 read-write / E2F→EZH2 writer	poised differentiation tripwire → G0 exit
APC/Cdh1 (C1) degrades EZH2 & Skp2 in G0	drug / input
CyclinD1 mitogen-dose (→EZH2, Skp2, growth)	